

UNIT III

INKJET TECHNOLOGY

Printer - Working Principle, Positioning System, Print head, Print bed, Frames, Motion control; Print head Considerations – Continuous Inkjet, Thermal Inkjet, Piezoelectric Drop-On-Demand; Material Formulation for jetting; Liquid based fabrication – Continuous jet, Multijet; Powder based fabrication – Colour jet.

Printer

The various different printing processes which we take for granted all involve the repeated reproduction of the same image or text many times. Usually, this is achieved by transferring a pattern of liquid or semi-liquid ink from some master pattern to the paper or other substrate through direct contact.

In contrast, the inkjet printer which is now ubiquitous in the modern home and office works on a fundamentally different principle. Each small droplet of ink, typically 10–100µm in diameter, is created and deposited under digital control, so that each pattern printed in a sequence can just as readily be different from the others as it can be the same.

What is Inkjet Technology?

Inkjet technology is a **non-contact printing process** that creates digital images by spraying tiny droplets of liquid ink directly onto a surface (paper, plastic, textiles, etc.). It is widely used in home and office printers, industrial printing, biomedical applications, and even 3D printing.

A wide range of materials can be deposited. By selection of an appropriate print-head, liquids with viscosities from 1 to 50mPas or higher can be printed. Several different methods can be used to generate printed structures.

Inkjet Printer – Working Principle

Uses liquid ink sprayed through tiny nozzles onto paper.

Steps Involved:

Data Processing

The computer sends the print command to the printer.

The printer driver translates the digital image into a printable format.

Ink Droplet Formation

Ink is stored in cartridges and released through nozzles.

Two types of inkjet printing mechanisms:

Thermal Inkjet: Uses heat to vaporize ink, creating a bubble that pushes droplets onto paper. (Used in HP, Canon printers)

Piezoelectric Inkjet: Uses vibrations from piezoelectric crystals to force ink out. (Used in Epson printers)

Ink Deposition & Paper Movement

The printer head moves **left to right**, depositing tiny ink droplets.

Paper feeds forward gradually for the next row of ink.

Drying & Final Output

Ink dries quickly, producing a **sharp image or text**.

The print is ejected, completing the process.

Laser Printer – Working Principle

Uses a **laser beam** and **toner powder** to print on paper.

Steps Involved:

Data Processing

The computer sends the document to the printer.

The printer's controller **converts digital data into a printable image**.

Laser Imaging on Drum

A **laser beam** scans across a rotating drum, **charging specific areas**.

The drum is coated with a light-sensitive material that holds an **electrostatic charge**.

Toner Application

Toner (powdered ink) is attracted to the charged areas of the drum.

The toner sticks to the drum in the pattern of the image or text.

Transfer to Paper

The **paper passes under the drum**, and toner is transferred.

A high-voltage charge **fixes the toner onto the paper**.

Fusing & Output

The paper passes through **heated rollers (fuser unit)**, melting and bonding the toner to the paper.

The final print is ejected from the printer.

Feature	Inkjet Printer	Laser Printer
Ink Type	Liquid Ink	Toner Powder
Mechanism	Nozzles spray ink	Laser & Electrostatic Charge
Print Speed	Slower	Faster
Print Quality	High, good for photos	Excellent for text
Cost	Cheaper upfront	Expensive but cost-efficient for bulk printing

Printer Positioning System: Overview & Working Principle

A **Printer Positioning System** refers to the mechanism that controls the precise movement of the **print head, paper, and other mechanical components** in a printer. This ensures accurate ink or toner deposition, resulting in high-quality prints.

Key Components of a Printer Positioning System

Print head Positioning System

Moves the print head horizontally across the paper.

Uses **stepper motors, guide rails, and encoders** for precise movement.

Paper Feed System

Controls the movement of the paper through the printer.

Uses **rollers, sensors, and stepper motors** to ensure correct alignment.

Optical Encoders

Detects the position of the print head and paper.

Provides feedback for **real-time adjustments** to avoid misalignment.

Stepper Motors & Servo Motors

Used for precise control of print head movement and paper feeding.

Stepper motors move in fixed steps, while **servo motors** adjust dynamically.

Control Electronics & Sensors

The **microcontroller or processor** interprets print data and controls motion.

Optical and mechanical sensors ensure accurate positioning.

Printer Positioning System Overview

The positioning system controls the **motion of the printhead, print bed, and frames** to ensure correct ink or material deposition. It consists of:

- ✓ **Printhead Movement (X-axis & Y-axis control)**
- ✓ **Print Bed Movement (Z-axis control in 3D printers)**
- ✓ **Frames & Support Structure**
- ✓ **Motion Control System (motors, sensors, and encoders)**

2. Print head

Definition:

The **print head** is the component that **releases ink (in 2D printers) or material (in 3D printers)** onto the substrate.

Types of Print heads:

Inkjet Print head (2D Printing)

Uses **thermal or piezoelectric technology** to spray tiny ink droplets.

Moves horizontally on a rail to cover the width of the paper.

Laser Print head (Laser Printers)

Uses a laser beam to create an electrostatic image on a drum, which then transfers toner to paper.

Extruder Print head (3D Printing)

Uses a **heated nozzle** to deposit layers of molten material (filament or resin).

Key Functions:

- ✓ Deposits ink or material with **high precision**
- ✓ Moves in a controlled manner across the printing area
- ✓ Works with **stepper motors & encoders** for accuracy

Print Bed

Definition:

The **print bed** is the surface on which the material (paper in 2D printing or a platform in 3D printing) is placed.

Types of Print Beds:

Paper Feed Tray (2D Printers) – Moves paper forward as needed.

Heated Print Bed (3D Printers) – Helps filament adhere properly.

Moving Print Bed – Used in some 3D printers where the bed moves in **Y or Z directions**.

Key Functions:

- ✓ Supports the printed material
- ✓ Moves to adjust position during printing
- ✓ Provides heat (in 3D printers) for adhesion

Frames & Structure

Definition:

The **frame** is the rigid structure that supports all moving parts in a printer.

Types of Frames:

Fixed Frame – Common in inkjet and laser printers.

Movable Frames (in 3D printers) – Adjusts the print bed and nozzle position.

Key Functions:

- ✓ Provides structural stability
- ✓ Houses mechanical & electrical components
- ✓ Ensures smooth movement of print head & bed

Motion Control System

Definition:

The **motion control system** ensures the precise movement of the **printhead, print bed, and paper feed system**.

Main Components:

Stepper Motors & Servo Motors – Control motion along **X, Y, and Z axes**

Linear Rails & Bearings – Guide smooth movement of the printhead and bed

Encoders & Sensors – Detect position for high accuracy

Belts & Pulleys – Transfer motion efficiently

How It Works:

The **controller** sends signals to stepper motors.

Motors move the **printhead (X, Y) and bed (Z, if needed)**.

Encoders & sensors provide feedback to adjust motion.

The system ensures **precise printing without misalignment**.

Print head Considerations:

Although much of this section also applies to CIJ, this discussion assumes that we are dealing with a DOD system.

Resolution is the inverse of the closest spacing at which individual spots are placed (i.e. spots per unit length) and may be different between the printing direction and the perpendicular direction.

Resolution is conventionally described in terms of dots per inch (dpi).

Usually there are several trade-offs to consider between printer specifications such as drop size, resolution, print speed, drop placement accuracy and print-head reliability.

Print-head manufacturing constraints may mean that the inter-nozzle gap is larger than the intended resolution perpendicular to the direction of printing. In that case, the required resolution can be achieved in a number of ways:

By tilting the print-head to the direction of motion: in combination with appropriate timing, this will enable spots to be printed closer together in a line perpendicular to the print direction;

By overprinting using the same print-head: that is, print one or more subsequent passes over the same area to 'fill in the gaps';

By using more than one print-head for the same area: that is, position one or more extra print-heads which interleave their spots to print at the required resolution.

Continuous Inkjet (CIJ)

In this method of printing a continuous jet of liquid is formed by forcing it under pressure through a nozzle. It will tend to break up into drops, driven by surface tension forces

Normally in a CIJ printer, this natural behaviour is controlled by imposing a disturbance on the jet so that the jet breaks up into a series of drops with well-controlled size and spacing.

As drops are produced continually, some means is needed to select those to be printed and to remove (and re-use) the drops not required for printing.

This means that CIJ systems tend to be more complex than other inkjet technologies as they require a method to produce drops, a method to select drops and a method to recover and control the liquid.

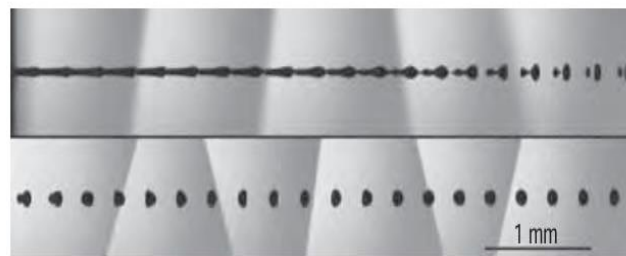
We shall consider both single-jet and multiple-jet variants.

Single-Jet CIJ:

One of the very first inkjet systems to be developed used a single continuous jet with electrostatic charging and deflection to select the drops and control their placement (Sweet, 1965).

The jet emerges from the nozzle with a velocity typically between 10 and 30ms⁻¹.

A regular disturbance on the jet imposed, for example, by the vibration of a piezoelectric structure causes the jet to break up into a stream of drops as shown in Figure



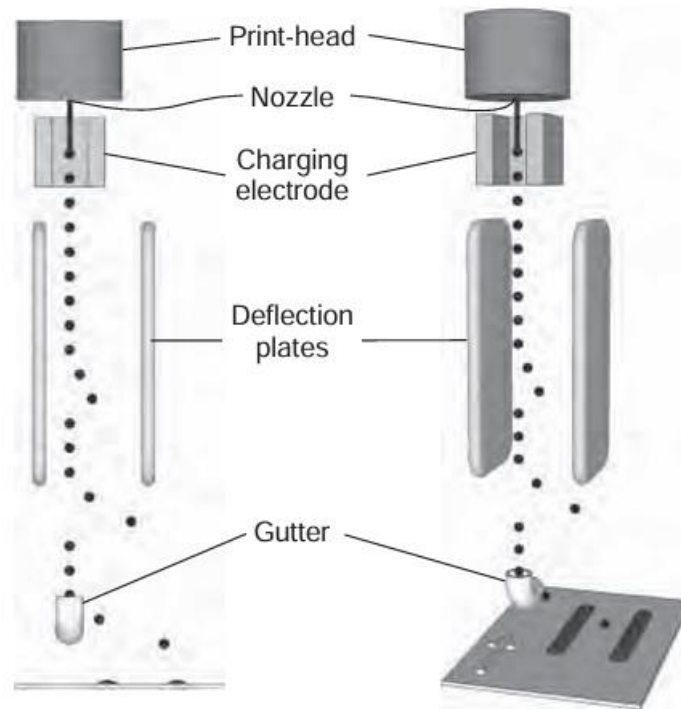
The break-up of a continuous jet to form drops (the lower image is a continuation of the upper image)

If not selected for printing, drops are collected in a catcher or gutter and the liquid is returned to the supply system for reconditioning and re-use.

To select drops, an electrode is placed close to the point where drops are forming from the continuous jet.

A potential on this electrode will induce an opposite electrical charge on the surface of the forming drop: for this to happen, the liquid must be sufficiently conductive.

When the drop breaks away, the charge is trapped on that particular drop and the potential of the electrode can be changed to put a different charge on the next drop (or set to zero for no charge).



Principles of operation of a single-jet continuous inkjet system

In this way, each successive drop can be charged as required. The selection is completed by causing the drop stream to move through a fixed electric field, maintained between two more electrodes, before reaching the catcher.

Multiple-Jet CIJ:

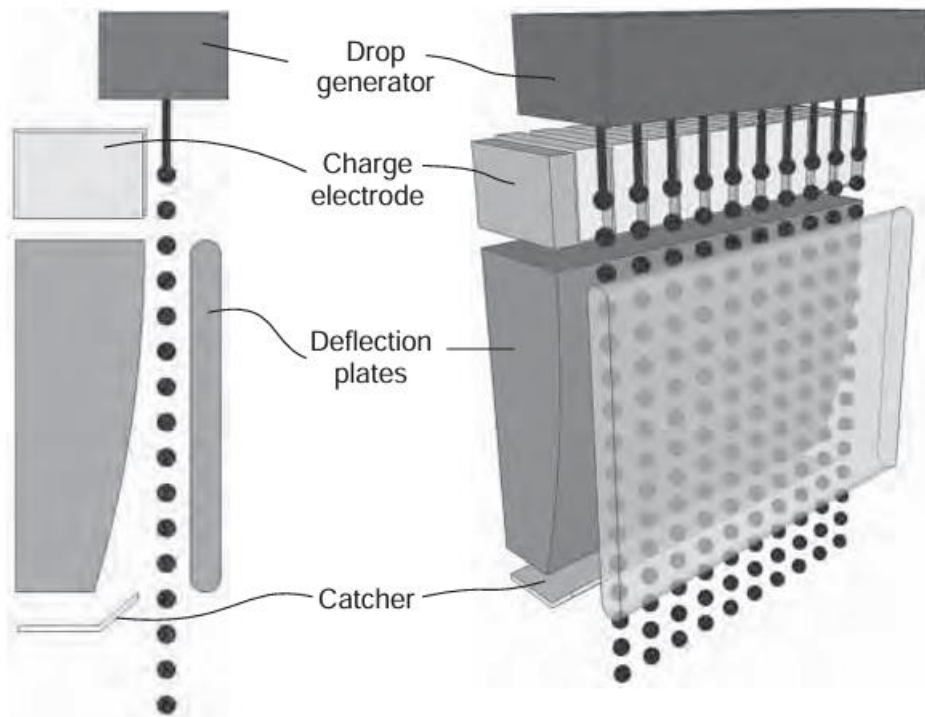
CIJ nozzles can be arranged in an array so that many jets or drop streams can be produced from a single print-head. Each jet is associated with its own charging electrode (or other means of drop selection), but the jets can share a common deflection electrode and drop catcher.

Normally each drop is either selected to print or not print and the substrate surface moves in a direction that is perpendicular to the jets but, in this case, parallel to the deflection field as shown in Figure. The print swath is determined by the width of the array.

An alternative means to impose a disturbance on the jet, used by Kodak in their recent multi-jet continuous jet systems (e.g. Hawkins and Pond, 2007), is to arrange a resistive heating element close to or at the nozzle.

This heater is modulated to change the jet temperature and hence the surface tension of the liquid. This in turn causes a flow which initiates a pinch point in the jet to form drops.

Kodak have explored alternative selection techniques, for example by using air flows to separate differently sized drops (which can be created from one stream using the thermal pinch-off method) or by switching the jet direction by using asymmetric heating with more complex heater arrangements at the nozzle.



Principles of operation of a multiple-jet continuous inkjet system.

Thermal DOD:

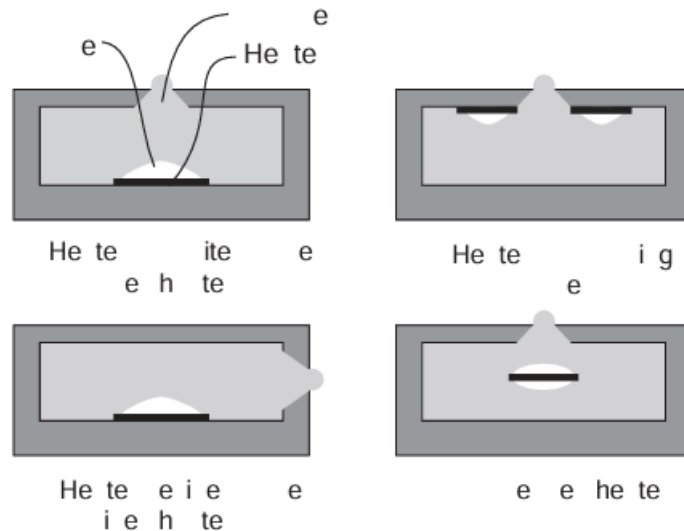
Thermal inkjet was invented and developed concurrently by Canon and Hewlett Packard in the late 1970s and early 1980s. This method is still used in a large proportion of home and office inkjet printers.

Within a cavity behind the nozzle is a small resistive heater in good thermal contact with the ink. Rapid heating (within a few micro-seconds) causes superheating and vaporisation of a thin layer of ink adjacent to the heater.

A vapour bubble quickly expands, producing the fluid displacement and energy necessary to force a drop from the nozzle. Once the drop is ejected and the heat source is switched off the bubble rapidly collapses, drawing in fresh ink, and the cycle is ready to start again once any residual pressure fluctuations and meniscus movements have died away.

One particular advantage of this technique is that the actuator is simply a resistive track which can be fabricated in a number of ways consistent with multi-layer fabrication processes such as photolithography.

- A wide variety of different thermal DOD designs have been described. Some of the differences concern the position of the heater relative to the nozzle.
- In face shooter designs, the heater is sited directly behind the nozzle as illustrated in (Figure a).
- The side shooter configuration has the heater on a wall beside the nozzle (Figure b).
- Other proposed designs include a heater on the nozzle face (Figure c) or
- suspended within the ink chamber (Figure d).



(A-d) Possible configurations of a drop-on-demand thermal inkjet print-head.

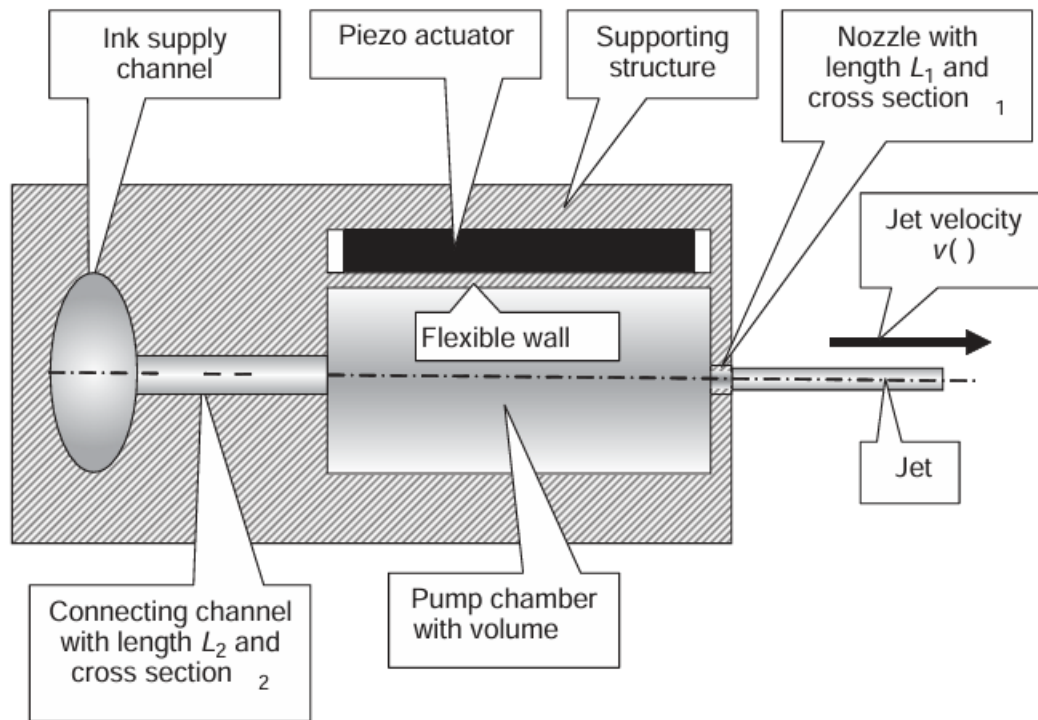
Piezoelectric DOD:

Drop ejection using PA was one of the first DOD techniques to be described (Zoltan, 1972; Kyser and Sears, 1976). PA-based inkjet print-heads are used in home and office inkjet systems, and PA has predominated in those print-heads developed for industrial application for either printing or digital fabrication.

Lead Zirconate Titanate (PZT) is a ceramic material that exhibits a strong piezoelectric effect and is used as the active component in many PA systems.

The piezoelectric effect is the ability of certain materials to generate an electric field in response to mechanical strain or, conversely, change shape in the presence of an electric field.

Several different actuation mechanisms have been developed. Push-mode actuators rely directly on the expansion and contraction of a PZT rod pushing a membrane, which changes the volume of an ink-filled reservoir behind a nozzle.



Basic arrangement of an inkjet printer pump. The volume of the pump chamber can be changed stepwise by charging the piezo actuator

Materials and Their Deposition by Inkjet Printing

Inkjet technology has been used to deposit a very wide range of materials, including metals, ceramics and polymers, for many different applications. Biological materials, including living cells, have also been successfully printed;

The most important restriction is that the substance being printed must be in liquid form (or contain small solid particles in a liquid medium) with appropriate rheological properties at the point of printing.

Deposition of Metals

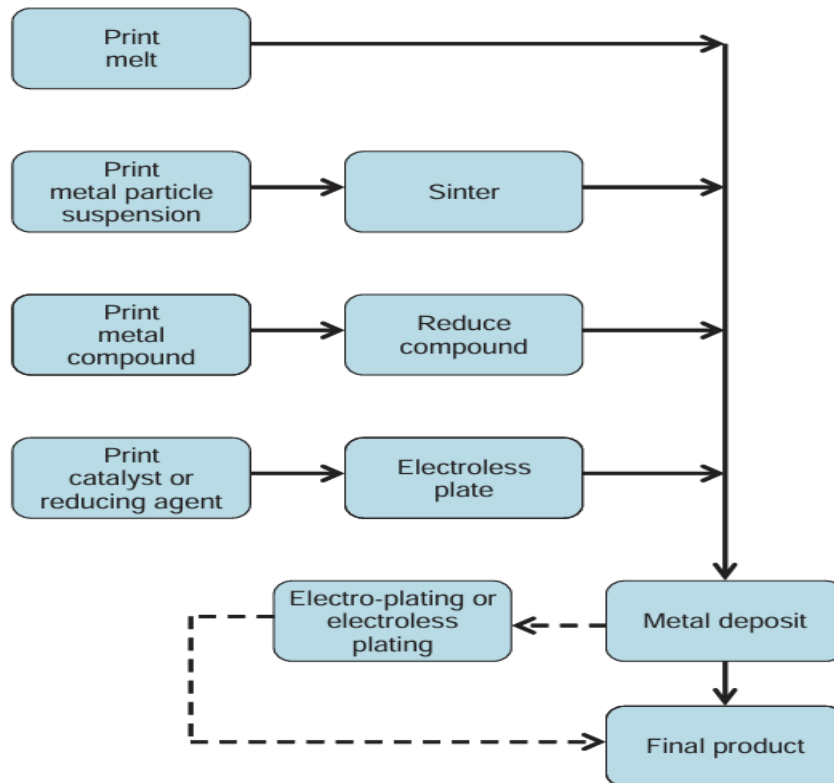
Several routes by which inkjet printing can be used to form metallic deposits.

These involve direct printing from a melt.

Printing a suspension of metallic particles which are then sintered to bond them together.

Printing a metal compound which is then chemically reduced to form the metal and printing a suitable catalyst followed by electro less plating to deposit the metal.

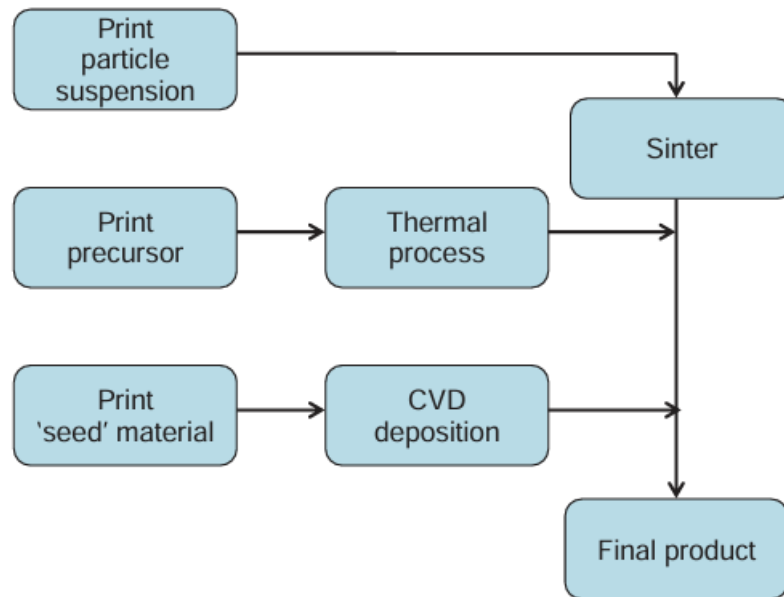
Any of these processes can also be combined with one or more secondary electro less plating or electro-plating steps to produce a thicker metallic deposit, which can even be of a different metal.



Deposition of Ceramics

There are several different routes by which ceramic materials can be deposited by inkjet printing, which are analogous to some of the methods used for metals and are illustrated in Figure

Suspensions of fine ceramic particles can be directly jetted, provided that the viscosity and surface tension of the mixture lie within the correct range. Alumina particles with median sizes between 0.3 and 1.5 μm have been mixed with wax and kerosene to produce jettable suspensions containing up to 40% ceramic particles by volume, which were then sintered after deposition to produce a material with a final relative density of 80%.

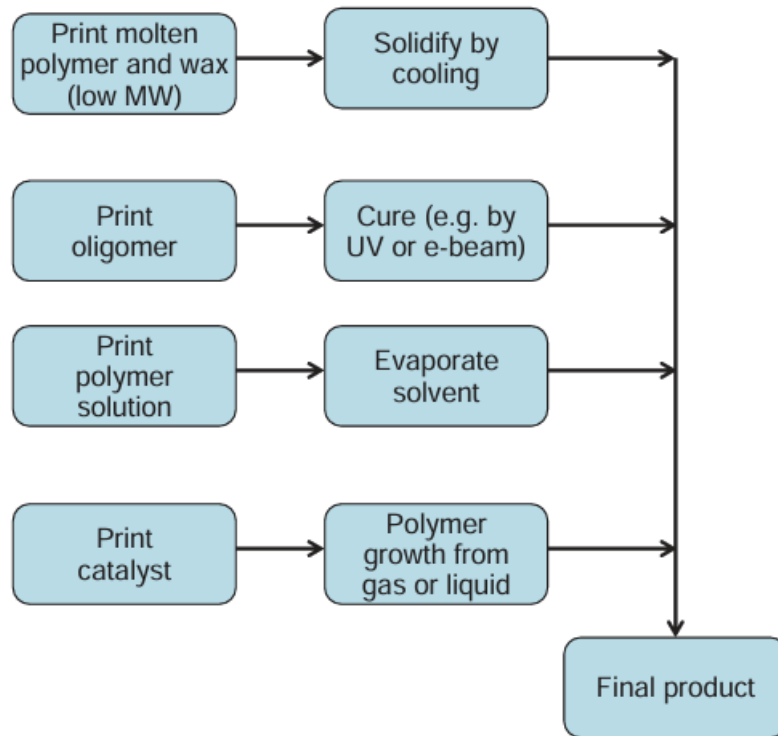


Deposition of Polymers

Summarises the methods by which polymeric materials can be deposited. Waxes, and other relatively short-chain polymers with molecular weights of a few hundred Daltons, can form readily jettable melts and can be used for some applications such as mask printing and rapid prototyping.

Long-chain polymers, however, cannot be jetted directly since even as a melt their viscosity is usually too great, and alternative routes are needed to deposit these polymers by inkjet printing.

Semiconductors and organic light-emitting diode (OLED) materials, can be printed in solution by inkjet methods. There is major commercial interest in printing organic semiconductors for such applications as display backplanes, and also in fabricating large-area polymer light-emitting diode (PLED) displays.



Liquid based fabrication – Continuous jet

A "liquid based fabrication continuous jet" refers to a manufacturing technique where a continuous stream of liquid material is precisely ejected from a nozzle, allowing for the deposition of droplets in a controlled pattern to build a 3D object, essentially a form of additive manufacturing that utilizes a continuous jet of liquid instead of a layer-by-layer approach; a prime example of this is "Continuous Inkjet Printing" (CIJ) where a steady stream of ink is precisely directed onto a surface to create printed text or images.

Key points about liquid based fabrication continuous jet:

Continuous flow:

Unlike some other 3D printing methods, the liquid is ejected in a continuous stream, enabling rapid deposition of material

Droplet formation:

The continuous jet is broken down into tiny droplets at the nozzle, which can be precisely controlled in terms of size and placement.

Curing mechanism:

Depending on the material, the droplets may solidify upon contact with a substrate or be cured using a secondary process like UV light.

High resolution potential:

Due to the precise control of droplets, this technique can achieve high resolution and detailed features

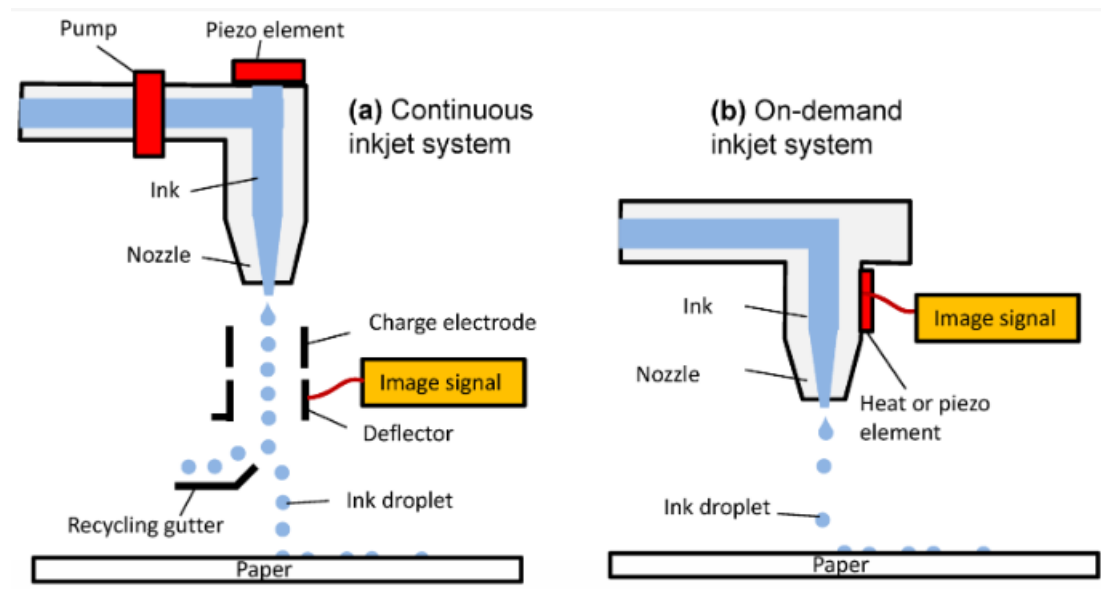
Applications of liquid based fabrication continuous jet:

Electronics printing: Deposition of conductive inks to create circuits on flexible substrates.

Bioprinting: Printing cells and biomaterials to create tissue constructs

Microfluidics: Creating complex microfluidic channels by precisely depositing different liquids

Packaging marking: Printing barcodes, dates, and other information on product packaging



Liquid based fabrication – Multi `jet

"Liquid based fabrication multi jet" refers to a 3D printing technology, most commonly known as "Multi Jet Fusion (MJF)," where a liquid agent is precisely deposited onto a powder bed using multiple tiny jets from a print head, essentially "drawing" the desired shape by selectively fusing the powder particles together to create a solid object; this process is characterized by high resolution, smooth surface finish, and the ability to print complex geometries without requiring support structures.

Key points about liquid based fabrication multi jet (MJF):

Powder bed process:

Unlike other 3D printing methods, MJF uses a bed of fine plastic powder (like nylon) as the building material.

Liquid fusing agent:

Instead of a laser, a print head with multiple nozzles precisely deposits a liquid "fusing agent" onto the powder bed, which selectively melts the powder particles where it lands.

High resolution and quality:

Due to the precise liquid deposition, MJF produces parts with excellent surface quality and fine details.

No support structures needed:

The design of the fusing agent allows for overhangs and complex geometries to be printed without the need for additional support structures.

Common applications of MJF:

Prototyping complex parts with high fidelity, Functional end-use parts requiring good mechanical properties and smooth surfaces, Medical devices, automotive components, and Aerospace parts

Powder-Based Fabrication Techniques:

This powder-based group of inkjet-based SFF systems utilizes inkjets to deliver liquid phase binders to aid selective consolidation of a powder bed.

Unlike liquid-based SFF systems, powder beds offer the added simplicity of not requiring support structures, as the surrounding, unbonded powder bed automatically supports the part being built.

They do, of course, require post-processing since the powder encasing the final object must be removed. This is almost like engaging in an archaeological dig, although system vendors provide enclosed glove-box units to make this operation relatively office friendly.

"Powder based fabrication - Colour jet" refers to a 3D printing technology called ColorJet Printing (CJP), where a thin layer of powder material is spread across a build platform and then selectively bonded together using a colored liquid binder deposited by an inkjet-like print head, allowing for full-color, detailed 3D models to be created layer by layer.

Key points about ColorJet Printing:

Process:

A thin layer of powder (the "core material") is spread, followed by the print head depositing a colored binder liquid in specific areas to solidify that part of the layer. The build platform then lowers, a new layer of powder is spread, and the process repeats.

Color capabilities:

By mixing different colored binders, the printer can achieve a wide range of colors, essentially producing a full-color 3D model.

Applications:

CJP is often used for creating detailed prototypes, models for visual communication, educational aids, and even low-volume production parts where color is critical